Mixed operations with integers

1 Evaluate:

a $-5+4+3$ **b** $-3+5-7$ **c** $6-(-3)+7$ **d** $-2-1-4$

e $1-(3+(-7))$ **f** $1-(4-(-5))$ **g** $((-8)+2)-5$ **h** $(4-(-2))-6$

2 Evaluate:

a $5+9\times(-6)$ **b** $(-4+2)\times 6$

c $-3-6\times(-4)$ **d** $(7-8)\times(-5)$

e $-5\times(-7)+8$ **f** $6\times(-4+(-7))$

g $6\times(-5)-4$ **h** $-8\times(-6-4)$

i $-6\times(-4)\times(-2)$ **j** $(2\times 27)-3\times 5-8$

3 Evaluate:

a $-7-((-8)\div 2)$ **b** $(-39+9)\div(-5)$

c $-9+28\div 4$ **d** $(21-6)\div(-3)$

e $48\div(-6)+(-4)$ **f** $14\div(-5+3)$

g $63\div(-9)-(-8)$ **h** $(-54)\div(-4-2)$

i $(-36)\div(-6)\div(-3)$ **j** $15\div(40\div(-8))$

4 Fill in the blank:

$$2+\square\times 5=-18$$

5 Evaluate:

a $-8+(24\div 4)^2$ **b** $-5+12\div 2^2$

c $(8-10\div 2)^2$ **d** $(-56\times 5)\div(-7)$

e $21\div(-7)\times 2$ **f** $(-84)\div(2\times 7)$

g $((-43+8)\div(-5))^2$ **h** $16\times(26\div(9+4))$

i $(48-((-9)\times 6)\div 3)+16$ **j** $((36-(10+10))\div 2)+14\times 6$

6 Explain why the following statement is incorrect, and find the correct answer:

$$6-(-60)\div(-3)=-22$$

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Applications

7 Three people on a farm are given five minutes to pick as many apricots as they can. They get to keep all the ripe apricots they pick, as well as an equal split of 30 ripe apricots between the three of them (which were picked earlier in the day).

If Justin (one of the three people) picks 33 apricots, 15 of which are unripe, how many apricots will Justin get to keep in total?

8 Each week Lisa gets paid an allowance based on her age and how many chores she did that week. The allowance consists of $\$1$ for each year old she is, $\$2$ for every chore she completes in a week, and is decreased by $\$4$ for each chore she forgets to complete.

If Lisa is 11 years old, completes 4 chores in a week, but forgets to complete a chore, how much will her allowance be that week?

9

a The temperature in Paris is 13 degrees Celsius less than double the temperature in London. If it is 8°C in London then what is the temperature in Paris?

b The temperature in Sydney is 21 degrees Celsius more than triple the temperature in New York. If it is -4°C in New York then what is the temperature in Sydney?

10 Fred busks with 2 friends whilst selling CD copies of his own music. They make a total of $\$36$ from busking and decide to split it evenly. Fred makes an additional amount by selling 9 CDs at $\$15$ each. If Fred pays a permit of $\$31$, what is the total amount he earns?

11 Laura and a group of 11 friends are ordering some pizza. They think that they will eat two pizzas for every three people. Pizzas cost $\$14$ each and there is a $\$8$ delivery fee.

If they split the total cost evenly between each person, how much will Laura need to pay?

12 At the start of the week, Quiana had $\$131$ in her bank account. On Monday, she withdrew $\$43$. On Wednesday, she deposited $\$11$, and on Thursday, she deposited three times as much as she deposited on Wednesday. Finally, she withdrew $\$25$ on Friday.

Find Quiana's final bank balance at the end of the week.

13 Starting from his house, Jerry walked $1\ \text{m/s}$ due south for 30 minutes, then $2\ \text{m/s}$ due north for 15 minutes and then, then due south again $2\ \text{m/s}$ for 45 minutes. Find Jerry's final distance from his house.

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